

# Control work 3 — ratio and proportion

Simplifying, sharing, chaining and proportion in one short paper.

LESSON 53

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 3

S7 12 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions

2 min

The paper

28 min

Answers and diagnosis

8 min

What comes next

2 min

LEARNING OBJECTIVES

- Simplify ratios, converting units first.
- Share a quantity in a two- and three-part ratio.
- Use direct and inverse proportion correctly.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 109–150

Cambridge

Stage 7, pp. 118–125

Workbook

Control paper 3

01

## Explanation

### The paper — 20 marks, 28 minutes

| Q | TASK                                                            | MARKS | FROM   |
|---|-----------------------------------------------------------------|-------|--------|
| 1 | Simplify $24 : 36$ and $75\text{ cm} : 3\text{ m}$              | 3     | L41    |
| 2 | In $3 : 5$ write each part as a fraction of the whole           | 3     | L42–43 |
| 3 | Share 480 in the ratio $5 : 7$                                  | 3     | L44–46 |
| 4 | Two shares are in the ratio $7 : 3$ and differ by 24: find both | 3     | L44–46 |
| 5 | Share 540 in the ratio $2 : 3 : 4$                              | 3     | L47–49 |
| 6 | 7 books cost 49 000: find the cost of 11                        | 3     | L50–52 |
| 7 | 5 workers take 12 days: how long for 10?                        | 2     | L50–52 |

THE ANSWERS

$2 : 3$  and  $1 : 4$ ;  $\frac{3}{8}$  and  $\frac{5}{8}$ ; 200 and 280; 42 and 18; 120, 180, 240; 77 000 sum; 6 days.

Naming the slip

| SLIP                                | WHAT IT LOOKS LIKE                  | THE FIX            |
|-------------------------------------|-------------------------------------|--------------------|
| units not converted                 | $75 : 3$                            | $75 : 300 = 1 : 4$ |
| denominator taken as the other part | $\frac{3}{5}$ of the whole          | $\frac{3}{8}$      |
| total divided by the wrong number   | $480 \div 5$                        | $480 \div 12$      |
| difference treated as the total     | $24 \div 10$                        | $24 \div 4$        |
| three-part total miscounted         | $2 + 3 + 4 = 8$                     | 9                  |
| inverse solved as direct            | 24 days                             | 6 days             |
| no check                            | shares that do not add to the total | add them back      |

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

| IF YOU LOST MARKS ON | IT RETURNS IN                                                     |
|----------------------|-------------------------------------------------------------------|
| Q1–Q2                | percentages, lesson 55 — a percentage is a ratio out of 100       |
| Q3–Q5                | the pie chart, lesson 160, where $360^\circ$ is shared in a ratio |
| Q6                   | speed, lesson 117 — distance is directly proportional to time     |
| Q7                   | every “more workers, fewer days” problem in the year              |

QUARTER I ENDS HERE

Directed numbers, algebra, the division of fractions and ratio are done. Quarter II opens with percentages, which uses all four.

## 02 Worked examples

**EXAMPLE 1** Model answer, Q4: two shares in ratio 7 : 3 differ by 24.

① The difference is  $7 - 3 = 4$  parts.  
Not 10.

②  $24 \div 4 = 6$  for one part.

③  $7 \cdot 6 = 42$  and  $3 \cdot 6 = 18$ .  
Check:  $42 - 18 = 24$  ✓

ANSWER

42 and 18

**EXAMPLE 2** Model answer, Q6: 7 books cost 49 000; find 11.

① Direct proportion.

② One book:  $49\,000 \div 7 = 7000$ .

③ Eleven:  $11 \cdot 7000 = 77\,000$ .

ANSWER

77 000 sum

**EXAMPLE 3** Model answer, Q7: 5 workers take 12 days; how long for 10?

① Inverse proportion — more workers, fewer days.

②  $5 \cdot 12 = 60$  worker-days.

③  $60 \div 10 = 6$  days.

ANSWER

6 days

**03** Interactive model

Return Q7 first and ask who doubled the days; naming the mistake as “solved an inverse problem as a direct one” makes it memorable.

**The block in eight questions**

24 : 36 simplifies to:

2 : 3

3 : 2

4 : 6

1 : 2

75 cm : 3 m simplifies to:

25 : 1

1 : 4

75 : 3

1 : 40

In 3 : 5 the first part is:

$\frac{3}{5}$

$\frac{3}{8}$

$\frac{5}{8}$

$\frac{5}{3}$

480 in 5 : 7 gives:

200, 280

240, 240

180, 300

220, 260

Ratio 7 : 3 with difference 24 gives one part:

2.4

4

6

8

540 in 2 : 3 : 4 gives:

120, 180, 240

100, 200, 240

135, 180, 225

90, 180, 270

7 books cost 49 000; 11 cost:

66 000

77 000

70 000

84 000

5 workers take 12 days; 10 take:

24

6

10

2.4

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | Ў‘ЗБЕКЧА                | РУССКИЙ                     |
|--------------------|-------------------------|-----------------------------|
| Control work       | Nazorat ishi            | Контрольная работа          |
| Ratio              | Nisbat                  | Отношение                   |
| To share           | Taqsimlash              | Разделить                   |
| One part           | Bir ulush               | Одна часть                  |
| Direct proportion  | To‘g‘ri proporsionallik | Прямая пропорциональность   |
| Inverse proportion | Teskari proporsionallik | Обратная пропорциональность |
| Units              | O‘lchov birligi         | Единицы                     |
| Diagnosis          | Tashxis                 | Диагностика                 |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Q1 requires first:

dividing

converting units

adding

multiplying

Q2 denominator is:

the other part

the total of the parts

the first part

10

Q3 divides 480 by:

5

7

12

35



Q4 divides 24 by:

10

7

3

4

Q6 is:

direct

inverse

neither

a ratio

Q7 is:

direct

inverse

neither

a ratio

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify  $24 : 36$

ANSWER  $2 : 3$

2. Simplify  $75 \text{ cm} : 3 \text{ m}$

ANSWER  $1 : 4$

3. In  $3 : 5$ , the first as a fraction

ANSWER  $\frac{3}{8}$

4. Share 480 in  $5 : 7$

ANSWER 200 and 280

5. Ratio  $7 : 3$ , difference 24

ANSWER 42 and 18

6. Share 540 in  $2 : 3 : 4$

ANSWER 120, 180, 240

7. 7 books cost 49 000: 11 books

ANSWER 77 000

Medium · the standard question

7 PROBLEMS

1. 5 workers take 12 days: 10 workers

ANSWER 6 days

2. Simplify 40 min : 2 h

ANSWER 1 : 3

3. Share 320 in 3 : 5

ANSWER 120 and 200

4. Ratio 5 : 2, difference 18

ANSWER 30 and 12

5. 9 m of cloth costs 72 000: 5 m

ANSWER 40 000

6. 6 taps fill a tank in 10 min: 4 taps

ANSWER 15 min

7. In 2 : 3 : 5, the last as a fraction

ANSWER  $\frac{1}{2}$

### Hard · stretch and reason

7 PROBLEMS

1.  $A : B = 2 : 3$ ,  $B : C = 4 : 5$ : find  $A : B : C$

ANSWER  $8 : 12 : 15$

2. Share 1 200 in  $1 : 2 : 3 : 6$

ANSWER 100, 200, 300, 600

3. Food for 20 people lasts 9 days: for 15 people

ANSWER 12 days

4. A ratio  $4 : 9$  with the total 91

ANSWER 28 and 63

5. Why is a ratio of  $75 : 3$  wrong for  $75 \text{ cm} : 3 \text{ m}$ ?

ANSWER The two numbers count different units

6. If 12 workers take 10 days, how many take 8?

ANSWER 15

7. Which of Q6 and Q7 keeps a constant product?

ANSWER Q7, the inverse one

## 07 Homework

### Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Simplify  $45 : 60$  and  $90 \text{ cm} : 2.7 \text{ m}$ .
3. Share 660 in the ratio  $4 : 7$ .
4. 8 pencils cost 24 000 sum. Find the cost of 14.
5. 6 workers take 15 days. How long would 9 take?